LAS (Legged Aerial Swarm): Requirements and Scope

1. Project Idea

In ideal circumstances, after natural disasters, we can send rescuers. But there are many situations where it may be dangerous for people to enter these areas to conduct search and rescue (SAR). In those cases, a quadruped robot acting as a mothership can carry a payload of autonomous drones in a swarm configuration to map and search the surroundings.

The quadruped and drone swarm together act as a scouting unit. Data is relayed back to the SAR team about dangers in the area, the locations of survivors, and a map of how the environment has changed.

2. Why This Helps SAR Teams

Search and rescue is inherently dangerous. Responders face severe weather, natural disasters, altitude, exposure, and psychological hazards.

LAS helps SAR teams identify survivors and dangers after the environment has changed. The quadruped can be sent into hard-to-reach locations, such as off-trail into dense forest, steep mountains, and unknown terrain. It deploys from a main junction like an access road or hiking trail, so responders can stay at a safe distance while it scouts ahead.

3. Primary Users and Decisions

Item	Definition
Primary user	SAR team lead or incident commander at a command post
Secondary users	Field responders, mapping and logistics staff
Decisions supported	Where to send rescuers first, which routes are safe, where hazards are, where survivors may be
What the user receives	A live-updating map, hazard markers, candidate survivor detections with location and imagery, and system status
Who confirms detections	A human operator, always

4. Primary Scenario for Version 1

The system covers several environments, and each needs a different design. For version 1, we focus on **one**:

Wilderness SAR deployed from a trailhead or access road (dense forest and rugged off-trail terrain).

Post-Natural Disaster Areas (Flooding, Earthquakes, Tornados, Hurricanes, etc.)

Environment	Main challenges	Version

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Dense forest and wilderness	Canopy blocks GPS and weakens radio, VIO struggles with foliage and low light, flying between trees is hard	**v1 target**
Steep or high-altitude terrain	Wind, cold-degraded batteries, thin air reduces drone lift, loose slopes challenge quadruped footing	Later
Large open fields	Long distances, wind exposure, longer drone range needed	Later
Collapsed or damaged structures	Tight spaces, dust, unstable debris, drones in voids	Later

5. Mission Concept

1. The team arrives at an access road, trail junction, or other staging point.
2. The operator defines the search area and mission goal.
3. The quadruped carries the drone payload toward the area, saving drone battery.
4. At launch points the robot selects, drones deploy for short sorties into areas the robot cannot reach.
5. Drones map, search, and flag possible survivors or hazards.
6. Data flows to the robot, which merges the map and relays it to the operator.
7. Drones return, dock, and recharge on the robot.
8. The operator reviews results and directs the next step.

6. Requirements

The numbers below are starting placeholders. They should be revised once the team agrees on the environment, budget, and hardware.

6.1 Performance targets

ID	Requirement	Placeholder target **[TBD]**
P-1	Area covered per mission	1 km² within 60 minutes
P-2	Drone sortie length	15 minutes, with a safety reserve
P-3	Autonomous launch-to-land success rate	95% or higher
P-4	Map accuracy versus ground truth	Within 0.5 m
P-5	Detection reporting latency	Under 60 seconds from detection to operator
P-6	Robot reach from the access point	2 km of trail or off-trail travel
P-7	Survivor detection performance	Detection rate and false-positive limit to be set after testing

ID	Requirement	Placeholder target **[TBD]**
P-8	Number of drones in v1	1 first, scaling to 3 or 4

6.2 Functional requirements

ID	Requirement
F-1	The system shall operate without GPS.
F-2	The robot shall build and maintain a shared map merged from robot and drone observations.
F-3	Drones shall launch, fly a sortie, return, land, and recharge on the robot without manual control.
F-4	The system shall flag candidate survivors and hazards and present them with location and imagery.
F-5	The system shall relay data to the operator and buffer it when the link is unavailable.
F-6	Drones shall complete or safely abort a sortie if the link to the robot is lost.
F-7	The system shall track team energy (robot and drones) and refuse missions it cannot complete safely.
F-8	The operator shall be able to pause, recall, or abort any drone or the whole mission at any time.

6.3 Safety and human control

ID	Requirement
S-1	A human operator shall confirm every survivor detection before action is taken.
S-2	The system shall have an emergency stop for the robot and a safety-pilot override for the drones.
S-3	Drones shall have defined failsafes: low battery, lost link, lost dock, and navigation failure.
S-4	The system shall not take any action that affects people on its own.

6.4 Environmental conditions (v1)

- Any conditions, as long as safe to operate.
- Operate during any time of day.
- Terrain limits for the robot (slope, obstacle height) based on the chosen platform.

7. Constraints and Assumptions

- No GPS is available. Cell coverage is limited or absent.
- Radio range is reduced by terrain and vegetation, so the robot acts as the relay.
- Drone battery life is short, which is the reason for the mothership design.
- The robot's battery must power travel, compute, comms, and drone charging.
- Version 1 starts with one drone before moving to a swarm.
- Landing happens only while the robot is stationary.
- Aviation, radio, and export-control rules apply to development and testing.

8. Out of Scope for Version 1

- Landing on a moving robot
- Night, heavy rain, or snow operation
- Operation inside collapsed structures
- Swarms larger than 3 or 4 drones
- Any autonomous action affecting people
- Military applications (see section 9)

9. Future Applications

Mapping technology can also support other uses. These are recorded here for the future and **do not drive version 1 requirements**:

- **Civilian:** surveying dense forests, mountains, large fields, and hard-to-reach terrain, for land management, conservation, and infrastructure inspection.
- **Research:** a testbed for legged-aerial teaming, multi-robot SLAM, and GPS-denied navigation.
- **Military and defense:** reconnaissance and mapping in hard-to-reach terrain. This direction brings export-control considerations (for example ITAR and EAR in the US) and stricter human-control policy. Decide early how this will be handled before sharing the project or collaborating outside the organization.

10. Success Definition

Version 1 succeeds if the system maps a defined wilderness area and reports a staged survivor to an operator, without GPS, in a field test.

11. Open Questions

1. Which environment is the v1 target, and who is the first test partner (a SAR team, a university, a training site)?
2. What is the budget, and does a robot or drone platform already exist?
3. What team and timeline are available?
4. Where will testing take place, and what permissions are needed?

12. Next Steps

1. Verify the robot platform's payload, power output, and SDK access before purchase.

2. Move to Phase 1: build the simulated sortie lifecycle.
3. Review legal and export-control requirements before field testing.